NAME:- SREEJA SANYAL

WEEK :-1

DAY:-5

DATE:-06.09.2026

Pipeline & Experiment Planning

1. Proposed Reproducible Pipeline

The complete Machine Learning workflow will be:

Battery Dataset

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Data Understanding & Exploration

↓

Data Cleaning

↓

Feature Selection

↓

Data Preprocessing

↓

Train/Validation/Test Split

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ML Model Training

↙ ↘

SOC Estimation Remaining Charging Time ↘ ↙

Model Evaluation& Comparison

↓

Best Performing Model

2. Data Preprocessing

The following steps are planned:

1. Load the raw dataset.
2. Preserve the original dataset without modification.
3. Create a separate working copy.
4. Inspect column names and data types.
5. Parse timestamps where required.
6. Check missing values.
7. Check invalid values.
8. Check duplicate observations.
9. Encode categorical variables such as charging status.
10. Scale features for models that require scaling.
11. Document all preprocessing operations.

3. Feature Engineering

Potential features include:

- Time/historical charging information
- Battery Voltage
- Charging Current
- Battery Temperature

· Charging power

Any feature that uses future information will not be used because it can cause data leakage.

4. Baseline Models

Linear Regression:-Linear Regression will be used as an interpretable baseline model. **5. Candidate Machine Learning Models**

· Decision Tree Regression:- Used to model nonlinear relationships. · Random Forest Regression:- Used as a stronger tree-based ensemble model. · Gradient Boosting:- Used as an advanced boosting-based regression method. · XGBoost:- May be used as an advanced gradient-boosting model if it is available in the project environment.

· ANN:-May be considered if the dataset is large enough and has meaningful sequential information.

6. Evaluation Metrics

For SOC and For Remaining Charging Time

- MAE
- RMSE
- R^2
- MSE

The two targets will be evaluated separately because they represent different quantities and units.

7. Proposed Repository Structure

- Battery-ML-Project
- ├── README.md
    - ├── data
        - ├── raw
        - ├── processed
        - └── README.md
- ├── notebooks
    - ├── 01_data_inspection.ipynb
    - ├── 02_data_preprocessing.ipynb
    - ├── 03_feature_engineering.ipynb
    - ├── 04_soc_models.ipynb
    - └── 05_rct_models.ipynb
- ├── src
    - ├── preprocessing
    - ├── features
    - ├── models
    - └── evaluation
- ├── Model_run
    - ├── Model_run _log.csv

- ├── 01_SOC_LinearRegression
- ├── 02_SOC_DecisionTree
- ├── 03_SOC_RandomForest ○

├── 04_SOC_GradientBoosting ○

├── 05_RCT_LinearRegression ○

├── 06_RCT_DecisionTree

- ├── 07_RCT_RandomForest ○

└── 08_RCT_GradientBoosting ○

- ├── models
  - ├── soc
  - └── rct
- |
- ├── results
  - ├── metrics
  - ├── plots
  - └── comparisons
- └── docs
  - ├── pipeline.md
  - ├── data_dictionary.md
  - ├── Model_run_plan.md
  - └── findings.md